WORKED SOLUTION

STEP III 1991, Question 5

Pure • Polar Curves and ODEs

STEP III 1991, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q5

The key idea

Forget the polar coordinates for a moment: the equation is just a linear second-order ODE with constant coefficients and θ as the variable. Solve it in the usual way (complementary function plus particular integral), fix the constants from the conditions at $\theta = \frac{\pi}{2}$, and then turn back to geometry. A curve given by $r = f(\theta)$ is a closed loop when r starts and finishes at 0 and stays positive in between, and its area is $\frac{1}{2}\int r^2 \,\mathrm{d}\theta$.

Solution

Solving the differential equation

The auxiliary equation $m^2 + 4 = 0$ has roots $m = \pm 2\mathrm{i}$, so the complementary function is

$$r = A\cos 2\theta + B\sin 2\theta.$$

For a particular integral, $\sin 3\theta$ is not in the complementary function, so we try $r = k\sin 3\theta$. Then $r'' = -9k\sin 3\theta$, and

$$-9k\sin 3\theta + 4k\sin 3\theta = 5\sin 3\theta \quad\Longrightarrow\quad k = -1.$$

So the general solution is

$$r = A\cos 2\theta + B\sin 2\theta - \sin 3\theta, \qquad \frac{\mathrm{d}r}{\mathrm{d}\theta} = -2A\sin 2\theta + 2B\cos 2\theta - 3\cos 3\theta.$$

At $\theta = \frac{\pi}{2}$ we have $\cos 2\theta = -1$, $\sin 2\theta = 0$, $\sin 3\theta = -1$ and $\cos 3\theta = 0$. The two conditions become

$$r = -A + 1 = 1 \quad\Longrightarrow\quad A = 0, \qquad \frac{\mathrm{d}r}{\mathrm{d}\theta} = -2B = -2 \quad\Longrightarrow\quad B = 1.$$

Hence

$$r = \sin 2\theta - \sin 3\theta.$$

The curve is a closed loop

To see where r vanishes and what sign it has, factorise with the identity $\sin P - \sin Q = 2\cos\frac{P+Q}{2}\sin\frac{P-Q}{2}$:

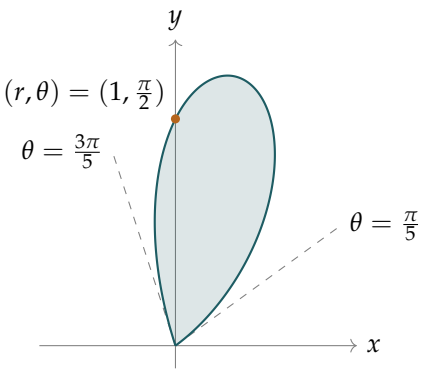
$$r = 2\cos\frac{5\theta}{2}\,\sin\left(-\frac{\theta}{2}\right) = -2\sin\frac{\theta}{2}\cos\frac{5\theta}{2}.$$

> **Ask yourself** Why does the factorised form settle the "closed loop" question?

Because both factors are easy to read off on the given range. For $\frac{\pi}{5} \le \theta \le \frac{3\pi}{5}$:

- $\frac{\theta}{2}$ lies in $\left[\frac{\pi}{10}, \frac{3\pi}{10}\right]$, so $\sin\frac{\theta}{2} > 0$;
- $\frac{5\theta}{2}$ lies in $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$, so $\cos\frac{5\theta}{2} \le 0$, with equality only at the two ends.

So $r = 0$ at $\theta = \frac{\pi}{5}$ and at $\theta = \frac{3\pi}{5}$, and $r > 0$ for every θ strictly between. The curve therefore leaves the origin along the ray $\theta = \frac{\pi}{5}$, sweeps round with positive r, and returns to the origin along the ray $\theta = \frac{3\pi}{5}$. Each ray $\theta = \text{const}$ in between meets the curve exactly once (at the single point with that $r > 0$), so the curve does not cross itself. It is a single closed loop, bounding the region $0 \le r \le \sin 2\theta - \sin 3\theta$, $\frac{\pi}{5} \le \theta \le \frac{3\pi}{5}$.



The area

The area enclosed is

$$\mathcal{A} = \frac{1}{2}\int_{\pi/5}^{3\pi/5} r^2\,\mathrm{d}\theta = \frac{1}{2}\int_{\pi/5}^{3\pi/5}\left(\sin^2 2\theta - 2\sin 2\theta\sin 3\theta + \sin^2 3\theta\right)\mathrm{d}\theta.$$

Using $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ and $2\sin P\sin Q = \cos(P-Q) - \cos(P+Q)$,

$$r^2 = 1 - \tfrac{1}{2}\cos 4\theta - \tfrac{1}{2}\cos 6\theta - \cos\theta + \cos 5\theta.$$

Write $D = \sin\frac{2\pi}{5} - \sin\frac{\pi}{5}$. We integrate term by term over $\left[\frac{\pi}{5}, \frac{3\pi}{5}\right]$, reducing each sine to $\pm\sin\frac{\pi}{5}$ or $\pm\sin\frac{2\pi}{5}$ using $\sin(x+2\pi) = \sin x$ and $\sin(\pi - x) = \sin x$:

$$\begin{aligned}
\int 1\,\mathrm{d}\theta &= \tfrac{2\pi}{5},\\
\int -\tfrac{1}{2}\cos 4\theta\,\mathrm{d}\theta &= -\tfrac{1}{8}\left[\sin\tfrac{12\pi}{5} - \sin\tfrac{4\pi}{5}\right] = -\tfrac{1}{8}\left[\sin\tfrac{2\pi}{5} - \sin\tfrac{\pi}{5}\right] = -\tfrac{1}{8}D,\\
\int -\tfrac{1}{2}\cos 6\theta\,\mathrm{d}\theta &= -\tfrac{1}{12}\left[\sin\tfrac{18\pi}{5} - \sin\tfrac{6\pi}{5}\right] = -\tfrac{1}{12}\left[-\sin\tfrac{2\pi}{5} + \sin\tfrac{\pi}{5}\right] = \tfrac{1}{12}D,\\
\int -\cos\theta\,\mathrm{d}\theta &= -\left[\sin\tfrac{3\pi}{5} - \sin\tfrac{\pi}{5}\right] = -D,\\
\int \cos 5\theta\,\mathrm{d}\theta &= \tfrac{1}{5}\left[\sin 3\pi - \sin\pi\right] = 0.
\end{aligned}$$

(For the third line, $\sin\frac{18\pi}{5} = \sin\left(-\frac{2\pi}{5}\right) = -\sin\frac{2\pi}{5}$ and $\sin\frac{6\pi}{5} = -\sin\frac{\pi}{5}$.) Adding,

$$\int_{\pi/5}^{3\pi/5} r^2\,\mathrm{d}\theta = \frac{2\pi}{5} + \left(-\frac{1}{8} + \frac{1}{12} - 1\right)D = \frac{2\pi}{5} - \frac{25}{24}D,$$

and halving,

$$\mathcal{A} = \frac{\pi}{5} - \frac{25}{48}D = \frac{\pi}{5} + \frac{25}{48}\left[\sin\frac{\pi}{5} - \sin\frac{2\pi}{5}\right],$$

which is the required result.

Checks

- At $\theta = \frac{\pi}{2}$: $r = \sin\pi - \sin\frac{3\pi}{2} = 1$ and $r' = 2\cos\pi - 3\cos\frac{3\pi}{2} = -2$, as required.
- Numerically, $\mathcal{A} \approx 0.6283 - \frac{25}{48}(0.3633) \approx 0.4391$. Simpson's rule on $\frac{1}{2}\int r^2\,\mathrm{d}\theta$, and the shoelace formula on the traced loop, both give 0.43911. Stepping the differential equation numerically from $\theta = \frac{\pi}{2}$ reproduces $r = \sin 2\theta - \sin 3\theta$ and confirms that r returns to 0 at both ends.
- The answer is plausible: the loop reaches $r \approx 1.22$ and spans an angle of $\frac{2\pi}{5}$. A sector of radius 1.22 and that angle has area about 0.93, and the loop fills a bit under half of it.

Common slips

- Getting the particular integral's sign wrong: $(-9+4)k = 5$ gives $k = -1$, not $+1$.
- Forgetting to show $r > 0$ between the endpoints. Two zeros of r alone don't make a loop: r could change sign in between, and then $\frac{1}{2}\int r^2\,\mathrm{d}\theta$ would not be the enclosed area.
- Reducing $\sin\frac{12\pi}{5}$ or $\sin\frac{18\pi}{5}$ carelessly: each is $\pm\sin\frac{2\pi}{5}$, and the sign matters.

Beyond STEP

The method here works because the equation is linear: every solution is a particular one plus a combination of the two independent solutions $\cos 2\theta$ and $\sin 2\theta$, and two initial conditions pick out exactly one solution. That is the existence and uniqueness theorem for linear ODEs, which first-year courses prove and use constantly. An attractive related fact is that $u = 1/r$ satisfies $u'' + u = \text{const}$ for orbits under an inverse-square force, which is how Newton's laws give conic-section orbits. Boyce and DiPrima, *Elementary Differential Equations*, covers the linear theory in detail.

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